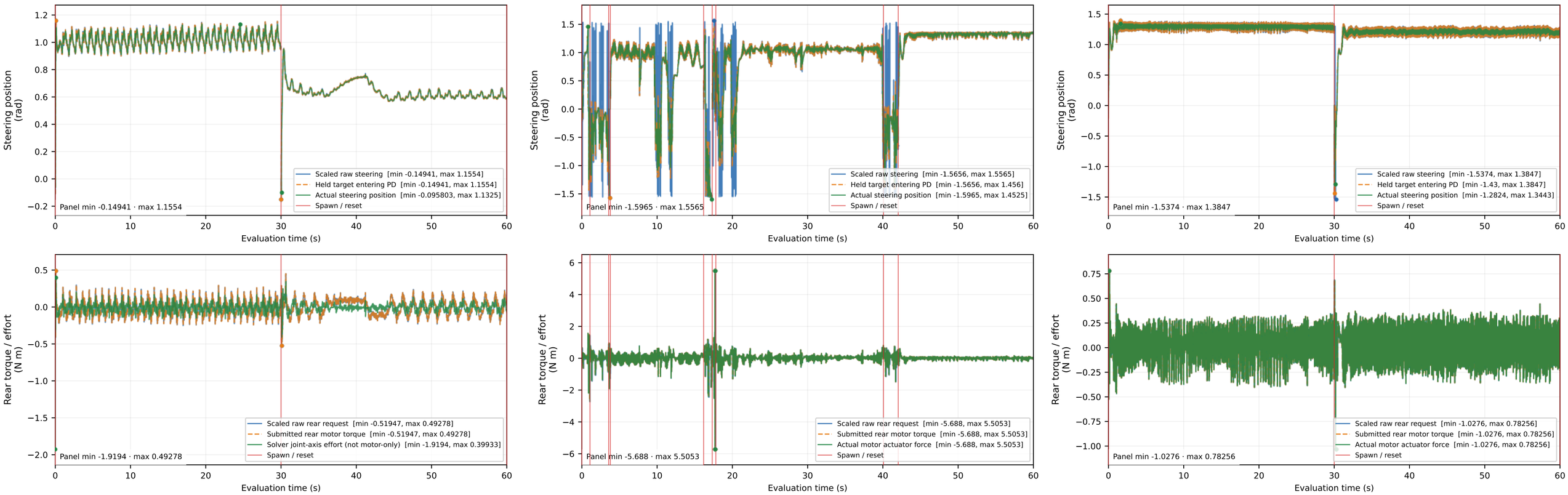
model_2500 · Demand versus actuation · env 16 · full 60 s



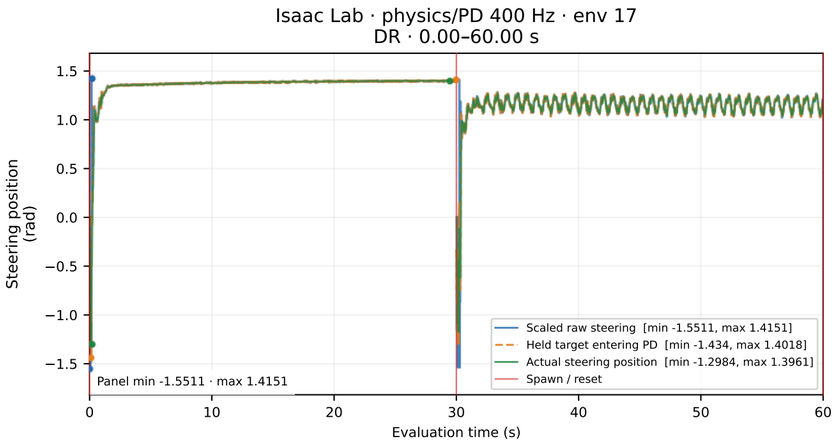
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

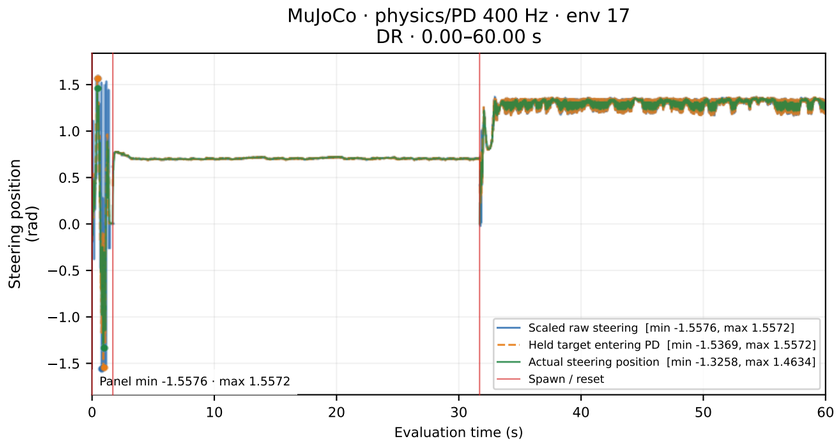
```
SAMPLED DR: env 16; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.891933, 0.959353, 0.992334, 0.924888]
  CoM XYZ mm [4.6868, 1.0248, -3.74794]
  mu_s [0.674963, 0.787792, 0.764681, 0.584424]
  mu_d [0.601294, 0.617134, 0.764681, 0.584424]
  Kp [steer,rear] [9.27673, 0]; Kd [0.181878, 0]
  spawn roll 0.0359022 rad; rate -0.236051 rad/s
  gyro bias XYZ rad/s [-0.000538585, 0.000225163, -0.00014116]
  gravity bias YZ [-0.00284693, 0.000550345]
E2 [30.00, 60.00) s:
  mass x [1.11595, 1.18185, 0.83303, 1.02334]
  CoM XYZ mm [2.93428, 4.99582, 0.202283]
  mu_s [0.579974, 0.760821, 0.764681, 0.673378]
  mu_d [0.579974, 0.670315, 0.764681, 0.673378]
  Kp [steer,rear] [9.60151, 0]; Kd [0.181605, 0]
  spawn roll 0.0172193 rad; rate -0.187324 rad/s
  gyro bias XYZ rad/s [0.000590245, 0.000248146, -0.000151491]
  gravity bias YZ [-0.000822571, -0.00160134]
Six significant digits shown; full precision: sampled_dr_values.json.
```

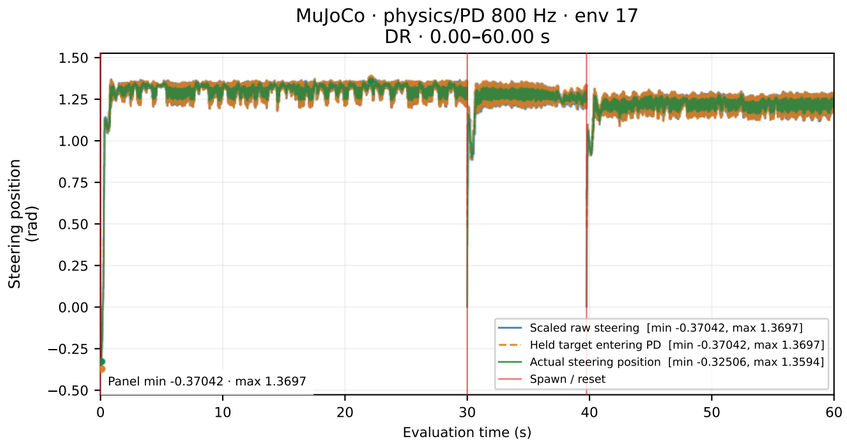
```
SAMPLED DR: env 16; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.08) s:
  mass x [0.923605, 1.15193, 0.897237, 0.883368]
  CoM XYZ mm [3.45813, -3.59427, -1.94667]
  mu_s [0.733515, 0.538434, 0.704749, 0.597748]
  mu_d [0.558392, 0.538434, 0.541926, 0.597748]
  Kp [steer,rear] [10.85, 0]; Kd [0.190802, 0]
  spawn roll -0.00998595 rad; rate -0.146724 rad/s
  gyro bias XYZ rad/s [-0.000146102, -1.28286e-05, -5.60525e-05]
  gravity bias YZ [-0.00196611, 0.000330421]
E2 [1.08, 3.56) s:
  mass x [1.14906, 0.916373, 0.94598, 0.994907]
  CoM XYZ mm [-3.77423, 1.72766, 0.690506]
  mu_s [0.597748, 0.631146, 0.749678, 0.723429]
  mu_d [0.597748, 0.631146, 0.741429, 0.723429]
  Kp [steer,rear] [9.3992, 0]; Kd [0.18431, 0]
  spawn roll -0.0585122 rad; rate -0.069853 rad/s
  gyro bias XYZ rad/s [0.000472192, -0.000347395, 0.000209527]
  gravity bias YZ [0.00233895, 0.000760862]
E3 [3.56, 3.78) s:
  mass x [0.872795, 1.07367, 0.898727, 1.09553]
  CoM XYZ mm [2.65337, 2.57532, -2.65282]
  mu_s [0.528253, 0.640867, 0.71008, 0.559972]
  mu_d [0.528253, 0.556841, 0.59371, 0.502209]
  Kp [steer,rear] [10.8385, 0]; Kd [0.197695, 0]
  spawn roll -0.0884651 rad; rate -0.212415 rad/s
  gyro bias XYZ rad/s [0.0002235, -0.000115595, 0.000163326]
  gravity bias YZ [-0.00208058, -0.00316361]
E4 [3.78, 16.18) s:
  mass x [0.995699, 0.958547, 1.15176, 0.98727]
  CoM XYZ mm [4.87388, 2.07726, -4.03266]
  mu_s [0.723429, 0.640016, 0.757579, 0.640016]
  mu_d [0.723429, 0.513141, 0.70921, 0.513141]
  Kp [steer,rear] [9.97118, 0]; Kd [0.186041, 0]
  spawn roll 0.0429654 rad; rate 0.0871774 rad/s
  gyro bias XYZ rad/s [-0.000137077, 0.00018035, 9.90635e-05]
  gravity bias YZ [0.000820886, -0.00174003]
E5 [16.18, 17.32) s:
  mass x [1.15268, 1.04228, 0.890194, 0.854027]
  CoM XYZ mm [-2.36818, -4.09797, 3.57273]
  mu_s [0.616244, 0.541939, 0.732187, 0.69316]
  mu_d [0.586498, 0.534359, 0.631664, 0.69316]
  Kp [steer,rear] [9.12976, 0]; Kd [0.193339, 0]
  spawn roll 0.0433519 rad; rate -0.12468 rad/s
  gyro bias XYZ rad/s [-2.57608e-05, 1.42294e-05, -8.98742e-05]
  gravity bias YZ [-0.00052632, 0.00179672]
E6 [17.32, 17.80) s:
  mass x [1.14067, 0.926093, 0.909646, 0.951796]
  CoM XYZ mm [1.46079, -3.27898, -3.3874]
  mu_s [0.736077, 0.559972, 0.631146, 0.666375]
  mu_d [0.699455, 0.502209, 0.631146, 0.519145]
  Kp [steer,rear] [9.61725, 0]; Kd [0.210762, 0]
  spawn roll -0.025371 rad; rate 0.172028 rad/s
  gyro bias XYZ rad/s [1.28074e-05, -0.000247523, 6.08333e-05]
  gravity bias YZ [0.00114787, -0.00239577]
E7 [17.80, 40.08) s:
  mass x [1.116, 1.01425, 0.882105, 1.02535]
  CoM XYZ mm [0.112943, 1.39924, -0.747668]
  mu_s [0.723429, 0.616244, 0.528253, 0.704749]
  mu_d [0.723429, 0.586498, 0.528253, 0.541926]
  Kp [steer,rear] [9.60372, 0]; Kd [0.210665, 0]
  spawn roll 0.0702297 rad; rate 0.0838734 rad/s
  gyro bias XYZ rad/s [-0.000403327, -0.000198575, 0.000226331]
  gravity bias YZ [0.00425945, 0.0019684]
E8 [40.08, 42.04) s:
  mass x [0.874414, 1.15853, 1.17124, 1.16893]
  CoM XYZ mm [0.943144, 2.07292, -3.16083]
  mu_s [0.538434, 0.728342, 0.71155, 0.71008]
  mu_d [0.538434, 0.728342, 0.71155, 0.59371]
  Kp [steer,rear] [10.6261, 0]; Kd [0.205094, 0]
  spawn roll -0.0572042 rad; rate -0.121939 rad/s
  gyro bias XYZ rad/s [7.89581e-05, -0.000117877, 0.000182835]
  gravity bias YZ [-0.00179059, 0.00136599]
E9 [42.04, 60.00) s:
  mass x [1.1668, 1.13148, 0.930705, 0.890804]
  CoM XYZ mm [3.5182, -4.51687, 4.93569]
  mu_s [0.640016, 0.71155, 0.637675, 0.546287]
  mu_d [0.513141, 0.71155, 0.637675, 0.546287]
  Kp [steer,rear] [9.47717, 0]; Kd [0.185056, 0]
  spawn roll -0.0297762 rad; rate 0.101724 rad/s
  gyro bias XYZ rad/s [-6.59488e-05, 0.000110389, -4.40996e-05]
  gravity bias YZ [-0.0016465, -0.00234535]
Six significant digits shown; full precision: sampled_dr_values.json.
```

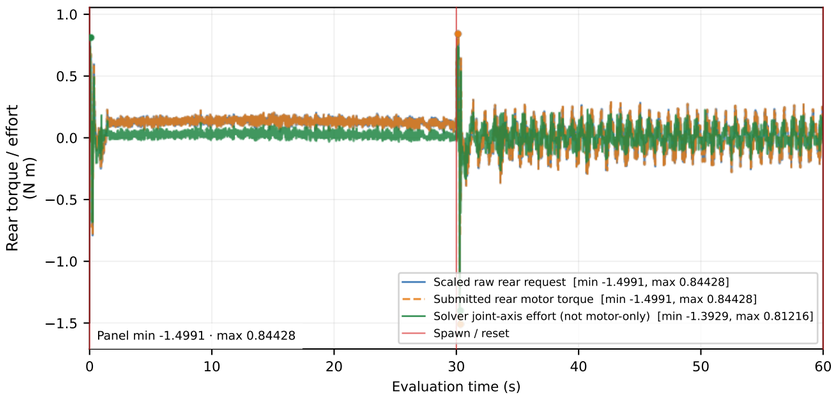
```
SAMPLED DR: env 16; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.923605, 1.15193, 0.897237, 0.883368]
  CoM XYZ mm [3.45813, -3.59427, -1.94667]
  mu_s [0.733515, 0.538434, 0.704749, 0.597748]
  mu_d [0.558392, 0.538434, 0.541926, 0.597748]
  Kp [steer,rear] [10.85, 0]; Kd [0.190802, 0]
  spawn roll -0.00998595 rad; rate -0.146724 rad/s
  gyro bias XYZ rad/s [-0.000146102, -1.28286e-05, -5.60525e-05]
  gravity bias YZ [-0.00196611, 0.000330421]
E2 [30.00, 60.00) s:
  mass x [0.808575, 0.946532, 0.921532, 0.807876]
  CoM XYZ mm [-4.19111, -1.08404, -4.92267]
  mu_s [0.538976, 0.637675, 0.640867, 0.633024]
  mu_d [0.538976, 0.637675, 0.556841, 0.568172]
  Kp [steer,rear] [9.44037, 0]; Kd [0.190149, 0]
  spawn roll -0.051795 rad; rate -0.166744 rad/s
  gyro bias XYZ rad/s [-7.02067e-05, -2.37253e-05, -0.000123674]
  gravity bias YZ [0.000558188, 0.0016771]
Six significant digits shown; full precision: sampled_dr_values.json.
```

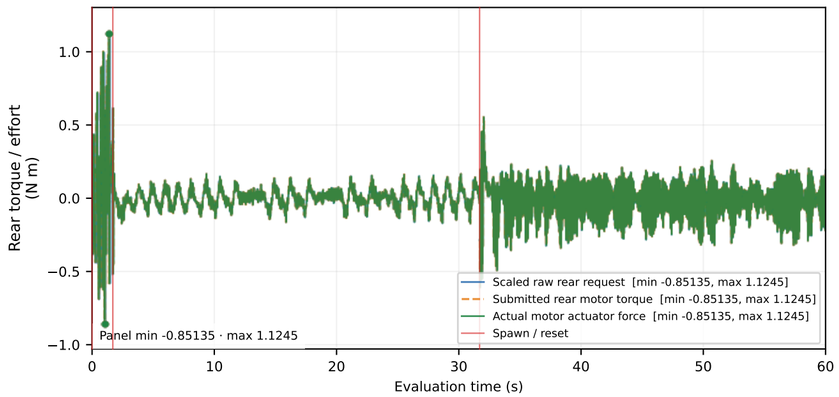
model_2500 · Demand versus actuation · env 17 · full 60 s

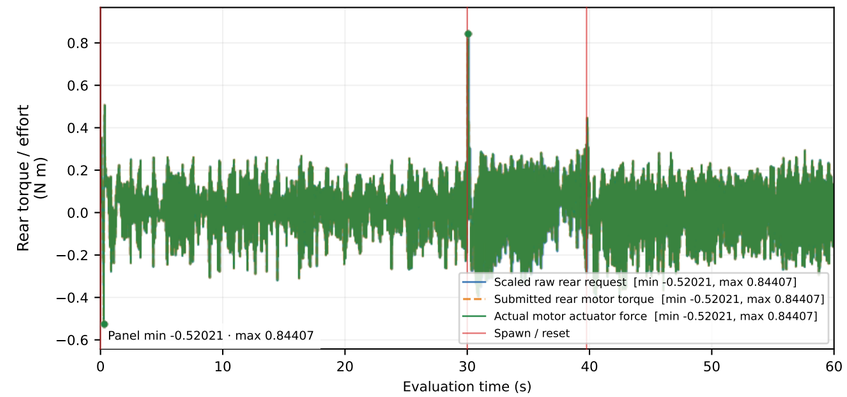












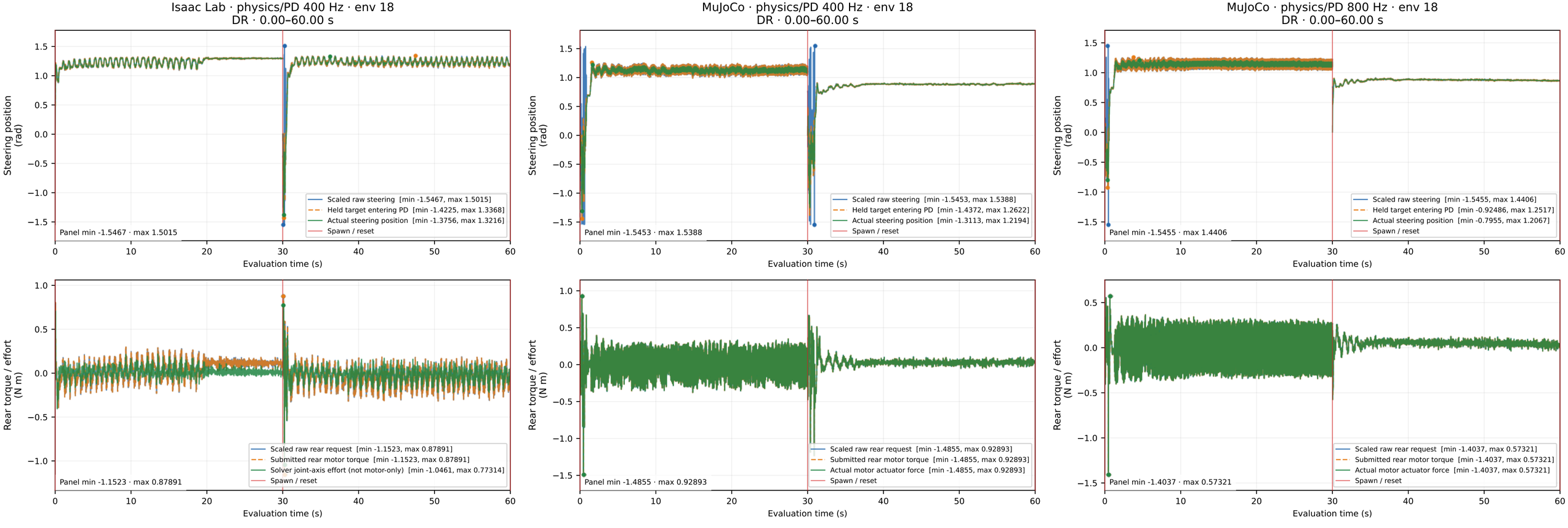
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 17; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.0488, 0.942439, 1.06045, 1.17219]
  CoM XYZ mm [-1.46499, -4.3528, -2.78383]
  mu_s [0.774409, 0.774409, 0.533455, 0.560535]
  mu_d [0.561095, 0.561095, 0.533455, 0.560535]
  Kp [steer,rear] [10.3908, 0]; Kd [0.217746, 0]
  spawn roll -0.0990896 rad; rate -0.0671556 rad/s
  gyro bias XYZ rad/s [-0.000199137, -0.000134969, 1.06368e-05]
  gravity bias YZ [-0.002044, 0.00241515]
E2 [30.00, 60.00) s:
  mass x [0.887591, 1.1975, 1.14532, 0.927279]
  CoM XYZ mm [3.31138, -0.607744, -3.82507]
  mu_s [0.659445, 0.532645, 0.592046, 0.673378]
  mu_d [0.547619, 0.532645, 0.534947, 0.673378]
  Kp [steer,rear] [10.0425, 0]; Kd [0.186075, 0]
  spawn roll -0.085215 rad; rate -0.0130568 rad/s
  gyro bias XYZ rad/s [0.000162379, -0.000128116, 0.000317264]
  gravity bias YZ [-0.00315221, -0.000589016]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 17; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.70) s:
  mass x [1.0351, 1.13043, 0.846778, 0.950972]
  CoM XYZ mm [-3.13738, -3.39717, -4.9418]
  mu_s [0.71155, 0.669571, 0.541939, 0.728342]
  mu_d [0.71155, 0.669571, 0.534359, 0.728342]
  Kp [steer,rear] [10.869, 0]; Kd [0.208257, 0]
  spawn roll -0.0307122 rad; rate -0.0746429 rad/s
  gyro bias XYZ rad/s [8.25675e-06, -0.00013759, 0.000263549]
  gravity bias YZ [-0.00103796, -0.00173413]
E2 [1.70, 31.70) s:
  mass x [1.11575, 0.893922, 1.11038, 1.1095]
  CoM XYZ mm [-0.73171, 4.99729, -4.17946]
  mu_s [0.71008, 0.700521, 0.732187, 0.538976]
  mu_d [0.59371, 0.641329, 0.631664, 0.538976]
  Kp [steer,rear] [10.0259, 0]; Kd [0.217319, 0]
  spawn roll 0.0435569 rad; rate 0.0392872 rad/s
  gyro bias XYZ rad/s [0.000111394, 0.000156834, -0.000151999]
  gravity bias YZ [-0.00336816, -0.00419074]
E3 [31.70, 60.00) s:
  mass x [1.19586, 1.07154, 1.13235, 0.975779]
  CoM XYZ mm [4.70814, -2.75462, 0.661321]
  mu_s [0.640016, 0.631146, 0.541939, 0.749678]
  mu_d [0.513141, 0.631146, 0.534359, 0.741429]
  Kp [steer,rear] [10.1648, 0]; Kd [0.195039, 0]
  spawn roll -0.033512 rad; rate 0.179082 rad/s
  gyro bias XYZ rad/s [-0.000297572, -2.38207e-06, -0.000210845]
  gravity bias YZ [0.00170795, 0.000186021]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 17; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0351, 1.13043, 0.846778, 0.950972]
  CoM XYZ mm [-3.13738, -3.39717, -4.9418]
  mu_s [0.71155, 0.669571, 0.541939, 0.728342]
  mu_d [0.71155, 0.669571, 0.534359, 0.728342]
  Kp [steer,rear] [10.869, 0]; Kd [0.208257, 0]
  spawn roll -0.0307122 rad; rate -0.0746429 rad/s
  gyro bias XYZ rad/s [8.25675e-06, -0.00013759, 0.000263549]
  gravity bias YZ [-0.00103796, -0.00173413]
E2 [30.00, 39.76) s:
  mass x [1.0484, 0.935286, 0.928736, 1.06293]
  CoM XYZ mm [3.31686, -4.22521, 0.275083]
  mu_s [0.749678, 0.748289, 0.559972, 0.568073]
  mu_d [0.741429, 0.689499, 0.502209, 0.568073]
  Kp [steer,rear] [9.73886, 0]; Kd [0.184541, 0]
  spawn roll 0.0737145 rad; rate 0.0203807 rad/s
  gyro bias XYZ rad/s [-0.000277662, -5.4727e-05, 0.000191924]
  gravity bias YZ [-0.000151406, 0.00473621]
E3 [39.76, 60.00) s:
  mass x [1.10051, 0.929118, 1.16593, 0.84389]
  CoM XYZ mm [-2.34865, -1.86845, 4.7607]
  mu_s [0.568073, 0.541939, 0.727426, 0.732187]
  mu_d [0.568073, 0.534359, 0.606358, 0.631664]
  Kp [steer,rear] [10.7868, 0]; Kd [0.202279, 0]
  spawn roll 0.0280855 rad; rate 0.122722 rad/s
  gyro bias XYZ rad/s [-0.000107871, 2.07815e-05, 0.00015181]
  gravity bias YZ [-0.0024167, 0.000327487]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 18 · full 60 s



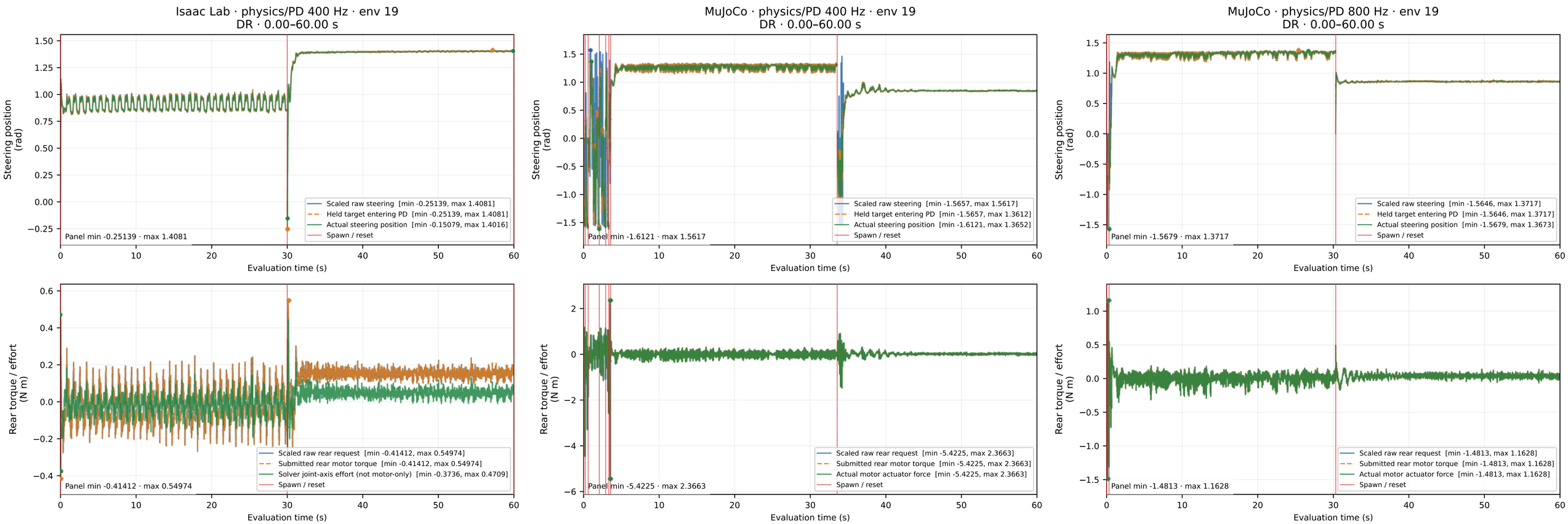
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 18; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.907865, 0.818429, 1.19701, 0.911529]
  CoM XYZ mm [-3.401, -1.14124, 3.49134]
  mu_s [0.760821, 0.576972, 0.674963, 0.632409]
  mu_d [0.670315, 0.576972, 0.601294, 0.589076]
  Kp [steer,rear] [10.1835, 0]; Kd [0.185575, 0]
  spawn roll 0.0792465 rad; rate -0.0982091 rad/s
  gyro bias XYZ rad/s [2.11285e-06, -0.00021168, 0.000204207]
  gravity bias YZ [0.00192337, -0.000994847]
E2 [30.00, 60.00) s:
  mass x [1.11078, 0.987952, 0.801457, 1.18803]
  CoM XYZ mm [3.67439, -1.87362, 0.500634]
  mu_s [0.659445, 0.659445, 0.592046, 0.760821]
  mu_d [0.547619, 0.547619, 0.534947, 0.670315]
  Kp [steer,rear] [9.01972, 0]; Kd [0.200358, 0]
  spawn roll -0.0920569 rad; rate -0.0218368 rad/s
  gyro bias XYZ rad/s [-0.000414288, -0.000261838, 7.58637e-05]
  gravity bias YZ [-0.00230514, 0.00243569]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 18; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.929714, 0.810859, 0.821888, 1.1962]
  CoM XYZ mm [3.6329, 0.505302, 4.68185]
  mu_s [0.637675, 0.669571, 0.757579, 0.733515]
  mu_d [0.637675, 0.669571, 0.70921, 0.558392]
  Kp [steer,rear] [9.63916, 0]; Kd [0.180825, 0]
  spawn roll -0.013749 rad; rate -0.194437 rad/s
  gyro bias XYZ rad/s [6.04374e-05, 0.00035961, 0.000141138]
  gravity bias YZ [-0.000964947, 0.00163817]
E2 [30.00, 60.00) s:
  mass x [1.1533, 1.07496, 0.981384, 1.16623]
  CoM XYZ mm [-4.98669, 2.89448, -0.758341]
  mu_s [0.538976, 0.728342, 0.748289, 0.546287]
  mu_d [0.538976, 0.728342, 0.689499, 0.546287]
  Kp [steer,rear] [10.7239, 0]; Kd [0.20776, 0]
  spawn roll -0.00389257 rad; rate 0.243496 rad/s
  gyro bias XYZ rad/s [0.000289537, 0.000290972, 0.000108021]
  gravity bias YZ [-0.00192329, 0.00103237]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 18; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.929714, 0.810859, 0.821888, 1.1962]
  CoM XYZ mm [3.6329, 0.505302, 4.68185]
  mu_s [0.637675, 0.669571, 0.757579, 0.733515]
  mu_d [0.637675, 0.669571, 0.70921, 0.558392]
  Kp [steer,rear] [9.63916, 0]; Kd [0.180825, 0]
  spawn roll -0.013749 rad; rate -0.194437 rad/s
  gyro bias XYZ rad/s [6.04374e-05, 0.00035961, 0.000141138]
  gravity bias YZ [-0.000964947, 0.00163817]
E2 [30.00, 60.00) s:
  mass x [1.1533, 1.07496, 0.981384, 1.16623]
  CoM XYZ mm [-4.98669, 2.89448, -0.758341]
  mu_s [0.538976, 0.728342, 0.748289, 0.546287]
  mu_d [0.538976, 0.728342, 0.689499, 0.546287]
  Kp [steer,rear] [10.7239, 0]; Kd [0.20776, 0]
  spawn roll -0.00389257 rad; rate 0.243496 rad/s
  gyro bias XYZ rad/s [0.000289537, 0.000290972, 0.000108021]
  gravity bias YZ [-0.00192329, 0.00103237]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 19 · full 60 s



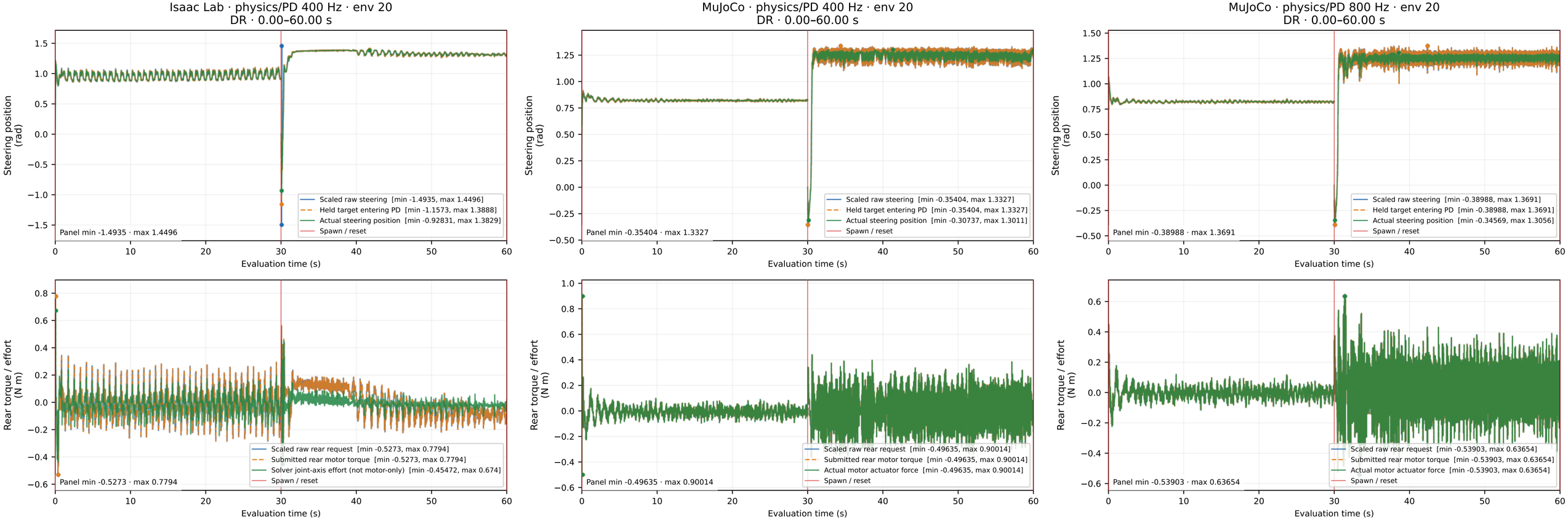
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 19; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.00429, 1.0378, 1.04659, 0.931652]
  CoM XYZ mm [2.20187, 2.93466, 3.4705]
  mu_s [0.598343, 0.703711, 0.576972, 0.764681]
  mu_d [0.598343, 0.703711, 0.576972, 0.764681]
  Kp [steer,rear] [10.741, 0]; Kd [0.205689, 0]
  spawn roll 0.0489928 rad; rate 0.230745 rad/s
  gyro bias XYZ rad/s [9.13323e-05, -7.36136e-05, -7.71893e-05]
  gravity bias YZ [0.00412858, 0.00170572]
E2 [30.00, 60.00) s:
  mass x [1.1418, 1.11129, 0.848825, 0.839723]
  CoM XYZ mm [-4.78759, -4.57539, -3.77113]
  mu_s [0.673378, 0.751971, 0.539956, 0.726175]
  mu_d [0.673378, 0.541224, 0.539956, 0.558574]
  Kp [steer,rear] [10.0199, 0]; Kd [0.219382, 0]
  spawn roll -0.00345242 rad; rate -0.178608 rad/s
  gyro bias XYZ rad/s [-0.00048902, -0.000157415, -0.000121871]
  gravity bias YZ [-0.0010147, 0.00292473]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 19; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [0.94975, 1.10877, 1.06846, 1.08366]
  CoM XYZ mm [-0.662472, 4.97269, 4.98093]
  mu_s [0.748289, 0.700521, 0.71155, 0.69316]
  mu_d [0.689499, 0.641329, 0.71155, 0.69316]
  Kp [steer,rear] [10.3709, 0]; Kd [0.204186, 0]
  spawn roll -0.0960328 rad; rate -0.238798 rad/s
  gyro bias XYZ rad/s [-0.000101158, 0.000235751, 0.000149081]
  gravity bias YZ [0.00322361, 0.00163551]
E2 [0.22, 0.60) s:
  mass x [1.16152, 1.09921, 0.911187, 0.870989]
  CoM XYZ mm [0.152926, 2.03583, -4.97262]
  mu_s [0.748289, 0.640867, 0.748289, 0.727426]
  mu_d [0.689499, 0.556841, 0.689499, 0.606358]
  Kp [steer,rear] [9.58862, 0]; Kd [0.205841, 0]
  spawn roll -0.0643534 rad; rate 0.0627738 rad/s
  gyro bias XYZ rad/s [0.000155158, -0.000431857, 0.000147435]
  gravity bias YZ [0.00131533, -0.0046838]
E3 [0.60, 2.08) s:
  mass x [0.859996, 0.850183, 1.05983, 0.888393]
  CoM XYZ mm [-4.74363, -4.76717, 4.30503]
  mu_s [0.749678, 0.700521, 0.748289, 0.757579]
  mu_d [0.741429, 0.641329, 0.689499, 0.70921]
  Kp [steer,rear] [10.3475, 0]; Kd [0.218515, 0]
  spawn roll -0.0411368 rad; rate 0.0241542 rad/s
  gyro bias XYZ rad/s [-1.22991e-05, -0.000179141, -3.93248e-06]
  gravity bias YZ [-0.00078878, 0.00196354]
E4 [2.08, 2.92) s:
  mass x [0.946687, 0.815144, 0.806943, 0.830702]
  CoM XYZ mm [0.369905, 0.356829, -3.35087]
  mu_s [0.791209, 0.666375, 0.666375, 0.666375]
  mu_d [0.767936, 0.519145, 0.519145, 0.519145]
  Kp [steer,rear] [9.02024, 0]; Kd [0.208556, 0]
  spawn roll -0.0149148 rad; rate 0.222933 rad/s
  gyro bias XYZ rad/s [0.000306687, -0.000102122, -3.7636e-05]
  gravity bias YZ [0.000331526, 0.000600647]
E5 [2.92, 3.34) s:
  mass x [1.02102, 0.991683, 0.990575, 0.808083]
  CoM XYZ mm [-1.19474, -2.0852, 2.67834]
  mu_s [0.611239, 0.749678, 0.528253, 0.616244]
  mu_d [0.611239, 0.741429, 0.528253, 0.586498]
  Kp [steer,rear] [10.9107, 0]; Kd [0.210466, 0]
  spawn roll -0.0600638 rad; rate 0.24504 rad/s
  gyro bias XYZ rad/s [-0.000331161, 0.000435587, -0.000245418]
  gravity bias YZ [0.00255999, 0.00126006]
E6 [3.34, 3.56) s:
  mass x [1.19377, 1.0168, 1.1573, 0.863469]
  CoM XYZ mm [2.70902, -3.40868, 1.48517]
  mu_s [0.616244, 0.631146, 0.640867, 0.69316]
  mu_d [0.586498, 0.631146, 0.556841, 0.69316]
  Kp [steer,rear] [9.02462, 0]; Kd [0.186443, 0]
  spawn roll 0.0919497 rad; rate 0.0209312 rad/s
  gyro bias XYZ rad/s [3.47464e-05, -6.53378e-05, -3.27807e-05]
  gravity bias YZ [-0.00232706, 0.00199089]
E7 [3.56, 33.56) s:
  mass x [0.96629, 1.15956, 1.12709, 1.14772]
  CoM XYZ mm [-1.28262, -1.86038, -1.8101]
  mu_s [0.538434, 0.736077, 0.538976, 0.71155]
  mu_d [0.538434, 0.699455, 0.538976, 0.71155]
  Kp [steer,rear] [10.4069, 0]; Kd [0.190008, 0]
  spawn roll 0.0358482 rad; rate -0.0162534 rad/s
  gyro bias XYZ rad/s [-2.00516e-05, -0.000129817, -0.000108534]
  gravity bias YZ [0.00326072, -0.00332642]
E8 [33.56, 60.00) s:
  mass x [0.936235, 1.02765, 0.988345, 1.15599]
  CoM XYZ mm [0.0885807, 3.90616, -3.79963]
  mu_s [0.528253, 0.611239, 0.568073, 0.559972]
  mu_d [0.528253, 0.611239, 0.568073, 0.502209]
  Kp [steer,rear] [10.8696, 0]; Kd [0.195431, 0]
  spawn roll -0.0335909 rad; rate 0.0919542 rad/s
  gyro bias XYZ rad/s [-0.000503023, 2.49212e-05, 0.000124039]
  gravity bias YZ [0.00275775, 0.00143926]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 19; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [0.94975, 1.10877, 1.06846, 1.08366]
  CoM XYZ mm [-0.662472, 4.97269, 4.98093]
  mu_s [0.748289, 0.700521, 0.71155, 0.69316]
  mu_d [0.689499, 0.641329, 0.71155, 0.69316]
  Kp [steer,rear] [10.3709, 0]; Kd [0.204186, 0]
  spawn roll -0.0960328 rad; rate -0.238798 rad/s
  gyro bias XYZ rad/s [-0.000101158, 0.000235751, 0.000149081]
  gravity bias YZ [0.00322361, 0.00163551]
E2 [0.32, 30.32) s:
  mass x [1.11003, 0.829622, 0.950311, 0.915982]
  CoM XYZ mm [-3.24813, -3.4527, 1.06287]
  mu_s [0.791209, 0.538976, 0.791209, 0.568073]
  mu_d [0.767936, 0.538976, 0.767936, 0.568073]
  Kp [steer,rear] [10.5427, 0]; Kd [0.19121, 0]
  spawn roll -0.0509555 rad; rate 0.0191255 rad/s
  gyro bias XYZ rad/s [-4.50528e-05, 6.58198e-05, -0.000161347]
  gravity bias YZ [0.00208725, -0.00132221]
E3 [30.32, 60.00) s:
  mass x [0.991628, 0.905446, 0.945454, 0.992245]
  CoM XYZ mm [-1.76291, 3.41365, -2.11742]
  mu_s [0.700521, 0.669571, 0.727426, 0.71155]
  mu_d [0.641329, 0.669571, 0.606358, 0.71155]
  Kp [steer,rear] [9.00862, 0]; Kd [0.188959, 0]
  spawn roll 0.0561029 rad; rate 0.159022 rad/s
  gyro bias XYZ rad/s [-0.000149479, 0.000295165, 7.51162e-05]
  gravity bias YZ [-0.000705262, -0.000130155]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 20 · full 60 s



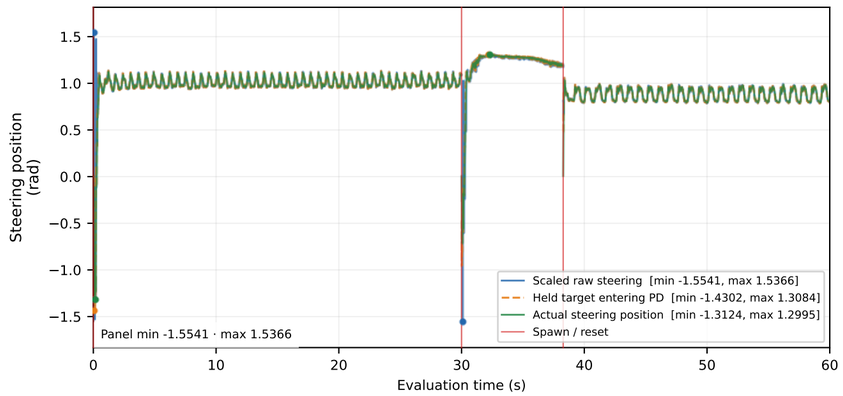
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

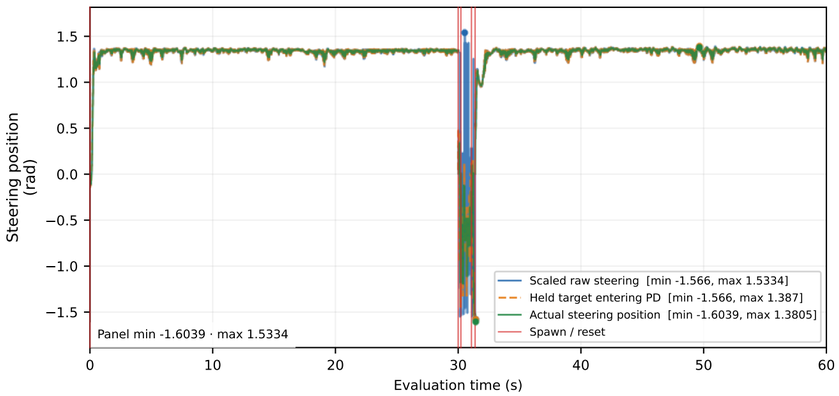
```
SAMPLED DR: env 20; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.12145, 0.998445, 1.07913, 1.05785]
  CoM XYZ mm [-1.91383, 1.48481, 4.09499]
  mu_s [0.576972, 0.760821, 0.713129, 0.602694]
  mu_d [0.576972, 0.670315, 0.713129, 0.602694]
  Kp [steer,rear] [9.81716, 0]; Kd [0.218603, 0]
  spawn roll 0.0737982 rad; rate 0.0786717 rad/s
  gyro bias XYZ rad/s [0.000357197, -0.000176057, -0.000173052]
  gravity bias YZ [-0.000308403, 0.000634569]
E2 [30.00, 60.00) s:
  mass x [0.892234, 0.823505, 0.947098, 0.964512]
  CoM XYZ mm [-0.711992, -3.47873, -4.83014]
  mu_s [0.576972, 0.726175, 0.547743, 0.785466]
  mu_d [0.576972, 0.558574, 0.547743, 0.52258]
  Kp [steer,rear] [9.22803, 0]; Kd [0.188455, 0]
  spawn roll -0.0530459 rad; rate 0.0152707 rad/s
  gyro bias XYZ rad/s [8.27099e-05, 0.000162087, 9.96022e-05]
  gravity bias YZ [0.00105849, -0.00303754]
Six significant digits shown; full precision: sampled_dr_values.json.
```

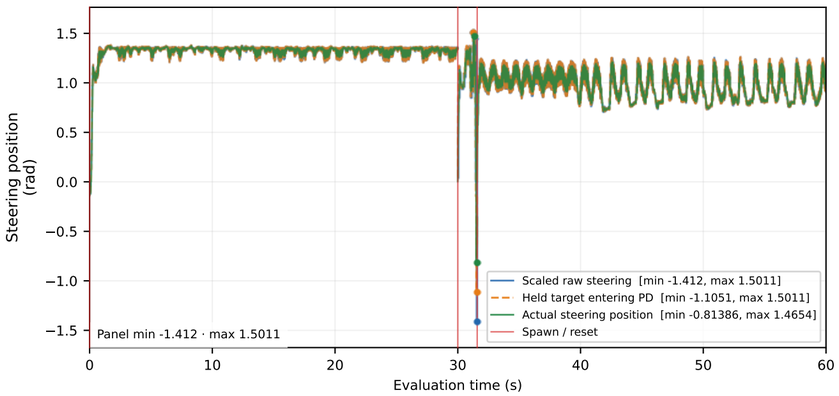
```
SAMPLED DR: env 20; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04373, 1.09911, 1.11228, 0.915142]
  CoM XYZ mm [1.93236, 4.26073, 0.13635]
  mu_s [0.541939, 0.71155, 0.727426, 0.736077]
  mu_d [0.534359, 0.71155, 0.606358, 0.699455]
  Kp [steer,rear] [9.47149, 0]; Kd [0.180846, 0]
  spawn roll 0.0663592 rad; rate 0.0689231 rad/s
  gyro bias XYZ rad/s [-7.44315e-05, 0.000269738, -0.000209089]
  gravity bias YZ [-0.000127877, 0.000950033]
E2 [30.00, 60.00) s:
  mass x [0.834144, 1.07257, 0.865316, 0.977668]
  CoM XYZ mm [-2.13668, -1.82604, -4.49653]
  mu_s [0.723429, 0.538434, 0.637675, 0.611239]
  mu_d [0.723429, 0.538434, 0.637675, 0.611239]
  Kp [steer,rear] [9.62454, 0]; Kd [0.191168, 0]
  spawn roll -0.0238987 rad; rate -0.10103 rad/s
  gyro bias XYZ rad/s [-3.95046e-05, 0.000102812, 0.000123087]
  gravity bias YZ [-2.13084e-05, 3.84637e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

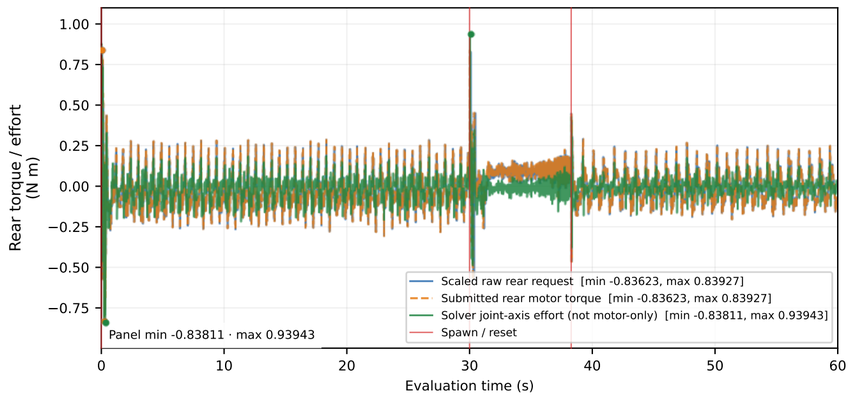
```
SAMPLED DR: env 20; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04373, 1.09911, 1.11228, 0.915142]
  CoM XYZ mm [1.93236, 4.26073, 0.13635]
  mu_s [0.541939, 0.71155, 0.727426, 0.736077]
  mu_d [0.534359, 0.71155, 0.606358, 0.699455]
  Kp [steer,rear] [9.47149, 0]; Kd [0.180846, 0]
  spawn roll 0.0663592 rad; rate 0.0689231 rad/s
  gyro bias XYZ rad/s [-7.44315e-05, 0.000269738, -0.000209089]
  gravity bias YZ [-0.000127877, 0.000950033]
E2 [30.00, 60.00) s:
  mass x [0.834144, 1.07257, 0.865316, 0.977668]
  CoM XYZ mm [-2.13668, -1.82604, -4.49653]
  mu_s [0.723429, 0.538434, 0.637675, 0.611239]
  mu_d [0.723429, 0.538434, 0.637675, 0.611239]
  Kp [steer,rear] [9.62454, 0]; Kd [0.191168, 0]
  spawn roll -0.0238987 rad; rate -0.10103 rad/s
  gyro bias XYZ rad/s [-3.95046e-05, 0.000102812, 0.000123087]
  gravity bias YZ [-2.13084e-05, 3.84637e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

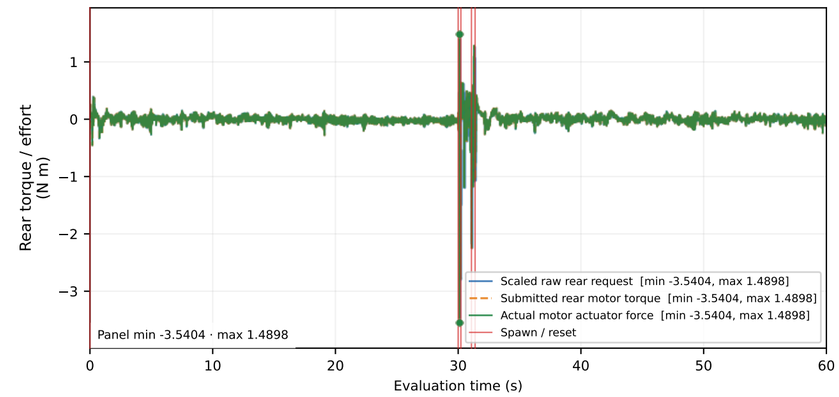
model_2500 · Demand versus actuation · env 21 · full 60 s

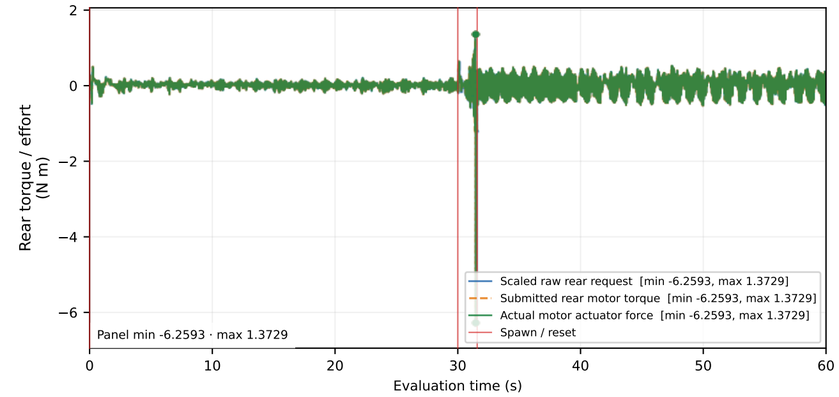












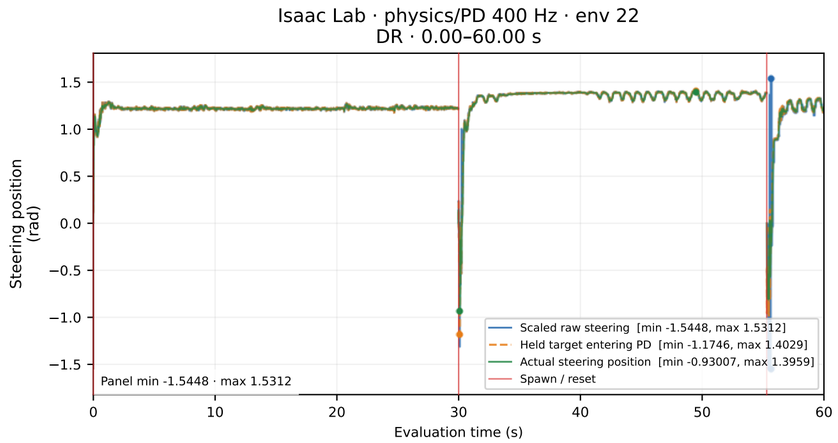
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

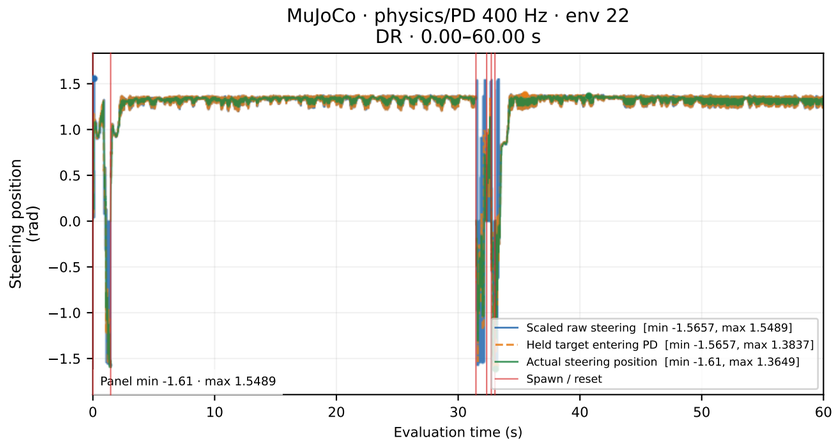
```
SAMPLED DR: env 21; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.16234, 0.931657, 0.874102, 0.941904]
  CoM XYZ mm [-0.778958, -0.191033, 4.88482]
  mu_s [0.576972, 0.764681, 0.760821, 0.532645]
  mu_d [0.576972, 0.764681, 0.670315, 0.532645]
  Kp [steer,rear] [10.558, 0]; Kd [0.190695, 0]
  spawn roll -0.0385937 rad; rate -0.201642 rad/s
  gyro bias XYZ rad/s [-0.000537161, -9.48584e-06, -0.000165546]
  gravity bias YZ [5.74883e-05, -0.000606045]
E2 [30.00, 38.28) s:
  mass x [1.07496, 1.12446, 0.947485, 1.11741]
  CoM XYZ mm [0.648886, -4.50441, 1.39657]
  mu_s [0.681711, 0.659445, 0.647391, 0.751971]
  mu_d [0.611756, 0.547619, 0.647391, 0.541224]
  Kp [steer,rear] [10.4772, 0]; Kd [0.21541, 0]
  spawn roll -0.0491127 rad; rate -0.225306 rad/s
  gyro bias XYZ rad/s [-9.14914e-05, 4.75103e-05, 3.46783e-05]
  gravity bias YZ [0.00205013, 0.00204799]
E3 [38.28, 60.00) s:
  mass x [0.925078, 0.848264, 1.02357, 1.1917]
  CoM XYZ mm [2.09817, 3.03209, -4.31511]
  mu_s [0.785466, 0.559809, 0.713129, 0.726175]
  mu_d [0.52258, 0.559809, 0.713129, 0.558574]
  Kp [steer,rear] [10.7923, 0]; Kd [0.180827, 0]
  spawn roll 0.0395315 rad; rate 0.215036 rad/s
  gyro bias XYZ rad/s [0.000233514, -4.39318e-05, 0.000166626]
  gravity bias YZ [-0.000890945, -0.00300513]
Six significant digits shown; full precision: sampled_dr_values.json.
```

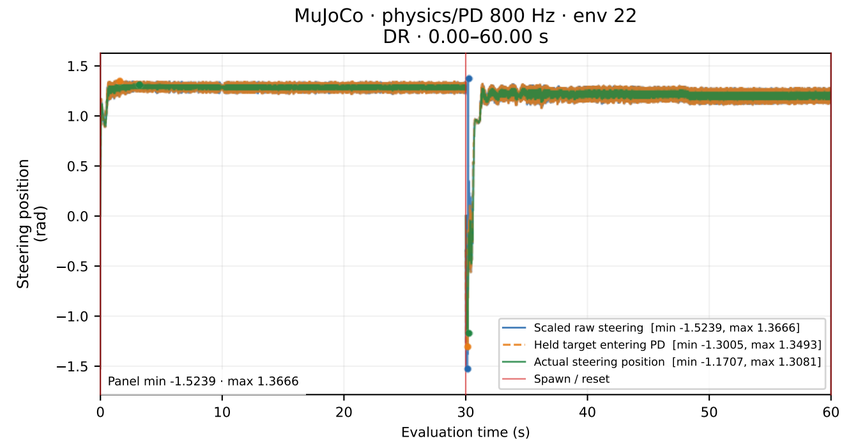
```
SAMPLED DR: env 21; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.02403, 1.1969, 0.878675, 1.00212]
  CoM XYZ mm [-2.99621, -3.27672, -1.3238]
  mu_s [0.538434, 0.748289, 0.727426, 0.640016]
  mu_d [0.538434, 0.689499, 0.606358, 0.513141]
  Kp [steer,rear] [10.979, 0]; Kd [0.199518, 0]
  spawn roll -0.0234954 rad; rate -0.0591556 rad/s
  gyro bias XYZ rad/s [-0.000390077, -5.04864e-05, -1.20618e-05]
  gravity bias YZ [0.00233172, -0.000743702]
E2 [30.00, 30.22) s:
  mass x [0.857328, 1.19008, 0.849359, 0.847076]
  CoM XYZ mm [2.84863, -3.96132, -1.49141]
  mu_s [0.631146, 0.538434, 0.749678, 0.700521]
  mu_d [0.631146, 0.538434, 0.741429, 0.641329]
  Kp [steer,rear] [9.09774, 0]; Kd [0.189967, 0]
  spawn roll 0.0231506 rad; rate -0.123269 rad/s
  gyro bias XYZ rad/s [7.24468e-05, -0.00027306, 0.000115335]
  gravity bias YZ [0.00107675, -0.00217639]
E3 [30.22, 31.08) s:
  mass x [1.151, 1.08475, 0.979946, 1.11803]
  CoM XYZ mm [-1.63771, 4.35002, 3.89226]
  mu_s [0.640016, 0.631146, 0.538434, 0.749678]
  mu_d [0.513141, 0.631146, 0.538434, 0.741429]
  Kp [steer,rear] [10.3954, 0]; Kd [0.209618, 0]
  spawn roll -0.0395964 rad; rate -0.196297 rad/s
  gyro bias XYZ rad/s [-0.000107801, -0.000248744, -0.00030249]
  gravity bias YZ [0.00149914, -0.00151463]
E4 [31.08, 31.38) s:
  mass x [1.02905, 1.06181, 1.19, 0.8162]
  CoM XYZ mm [4.10381, 4.61135, -0.571818]
  mu_s [0.71155, 0.559972, 0.631146, 0.541939]
  mu_d [0.71155, 0.502209, 0.631146, 0.534359]
  Kp [steer,rear] [9.43782, 0]; Kd [0.200375, 0]
  spawn roll -0.0668429 rad; rate 0.139502 rad/s
  gyro bias XYZ rad/s [-9.08241e-05, 1.6222e-05, -0.000179747]
  gravity bias YZ [-8.45339e-05, 0.000930439]
E5 [31.38, 60.00) s:
  mass x [1.10165, 0.913317, 0.839364, 1.16837]
  CoM XYZ mm [-3.28761, -3.97198, -0.854517]
  mu_s [0.597748, 0.616244, 0.538976, 0.538976]
  mu_d [0.597748, 0.586498, 0.538976, 0.538976]
  Kp [steer,rear] [10.3977, 0]; Kd [0.184072, 0]
  spawn roll -0.0282585 rad; rate 0.221166 rad/s
  gyro bias XYZ rad/s [-0.000123733, -3.5891e-05, 0.00017009]
  gravity bias YZ [-0.00179086, 0.000860961]
Six significant digits shown; full precision: sampled_dr_values.json.
```

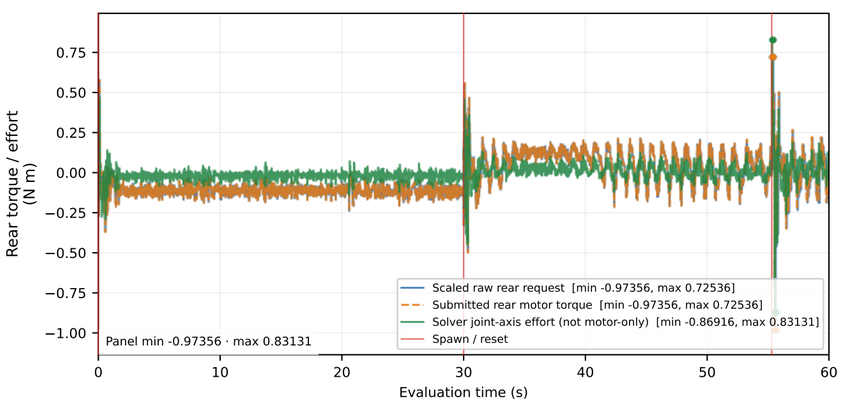
```
SAMPLED DR: env 21; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.02403, 1.1969, 0.878675, 1.00212]
  CoM XYZ mm [-2.99621, -3.27672, -1.3238]
  mu_s [0.538434, 0.748289, 0.727426, 0.640016]
  mu_d [0.538434, 0.689499, 0.606358, 0.513141]
  Kp [steer,rear] [10.979, 0]; Kd [0.199518, 0]
  spawn roll -0.0234954 rad; rate -0.0591556 rad/s
  gyro bias XYZ rad/s [-0.000390077, -5.04864e-05, -1.20618e-05]
  gravity bias YZ [0.00233172, -0.000743702]
E2 [30.00, 31.58) s:
  mass x [0.857328, 1.19008, 0.849359, 0.847076]
  CoM XYZ mm [2.84863, -3.96132, -1.49141]
  mu_s [0.631146, 0.538434, 0.749678, 0.700521]
  mu_d [0.631146, 0.538434, 0.741429, 0.641329]
  Kp [steer,rear] [9.09774, 0]; Kd [0.189967, 0]
  spawn roll 0.0231506 rad; rate -0.123269 rad/s
  gyro bias XYZ rad/s [7.24468e-05, -0.00027306, 0.000115335]
  gravity bias YZ [0.00107675, -0.00217639]
E3 [31.58, 60.00) s:
  mass x [0.824134, 0.974825, 1.1904, 0.828339]
  CoM XYZ mm [2.50539, 2.62759, -4.6295]
  mu_s [0.704749, 0.727426, 0.69316, 0.559972]
  mu_d [0.541926, 0.606358, 0.69316, 0.502209]
  Kp [steer,rear] [10.9743, 0]; Kd [0.184708, 0]
  spawn roll 0.0205726 rad; rate 0.0219324 rad/s
  gyro bias XYZ rad/s [-0.000251298, 0.000132695, 5.19054e-05]
  gravity bias YZ [0.000527718, -0.00169607]
Six significant digits shown; full precision: sampled_dr_values.json.
```

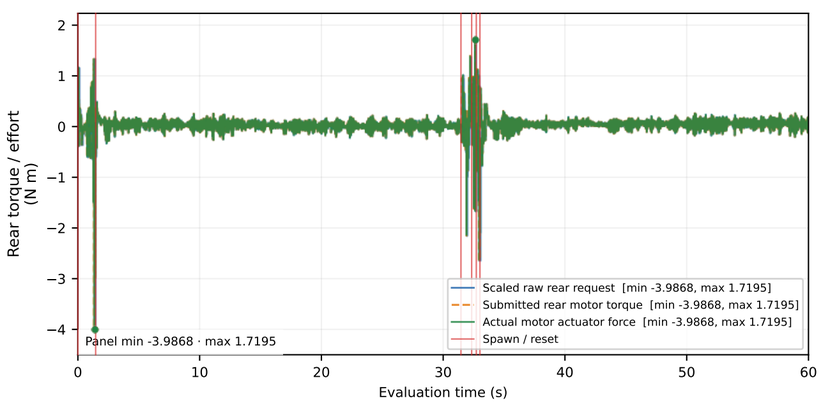
model_2500 · Demand versus actuation · env 22 · full 60 s

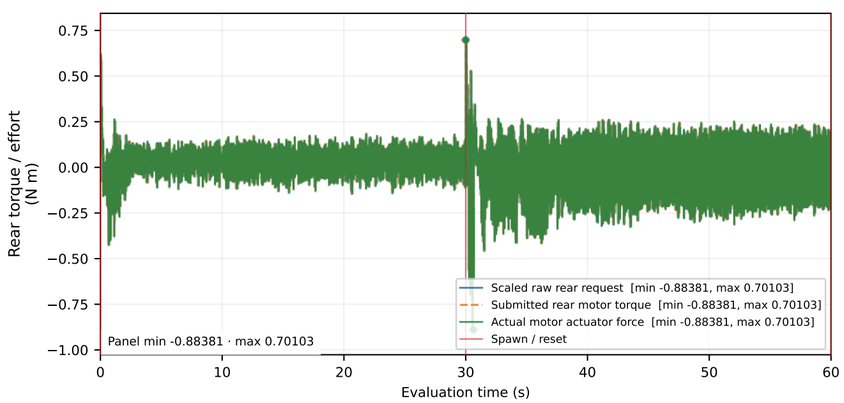












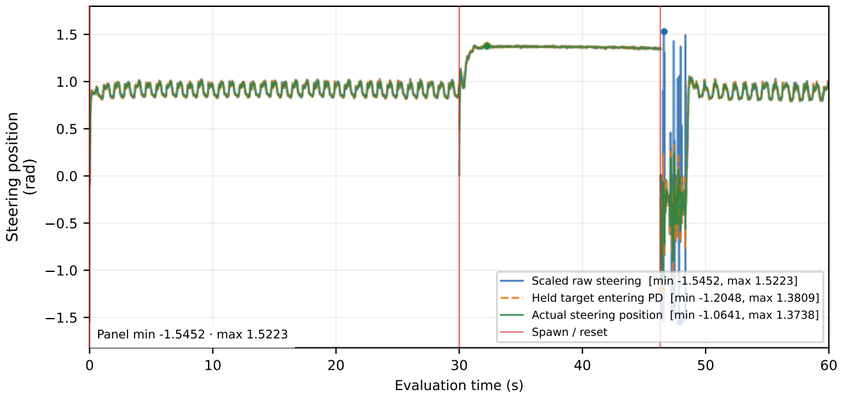
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

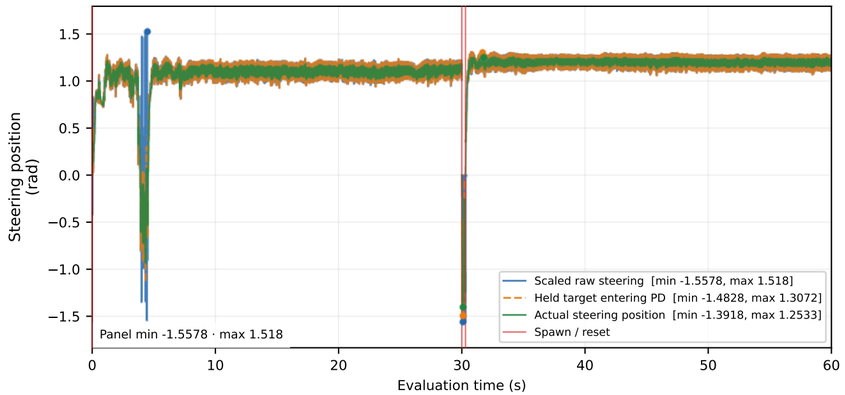
```
SAMPLED DR: env 22; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.09989, 0.913112, 1.11555, 1.16651]
  CoM XYZ mm [3.35515, -2.84438, 2.06447]
  mu_s [0.774409, 0.681711, 0.602694, 0.579974]
  mu_d [0.561095, 0.611756, 0.602694, 0.579974]
  Kp [steer,rear] [9.75244, 0]; Kd [0.205282, 0]
  spawn roll 0.0130913 rad; rate -0.061652 rad/s
  gyro bias XYZ rad/s [0.000563281, -0.000617255, 0.000169907]
  gravity bias YZ [0.0011242, 6.85766e-05]
E2 [30.00, 55.30) s:
  mass x [1.19175, 1.10707, 1.14586, 0.845834]
  CoM XYZ mm [-2.34434, -3.65275, -1.65352]
  mu_s [0.774409, 0.659445, 0.532645, 0.533455]
  mu_d [0.561095, 0.547619, 0.532645, 0.533455]
  Kp [steer,rear] [10.0599, 0]; Kd [0.215464, 0]
  spawn roll -0.0596136 rad; rate 0.198514 rad/s
  gyro bias XYZ rad/s [-4.66532e-05, -9.9831e-05, -0.000144437]
  gravity bias YZ [0.00113814, 0.00285899]
E3 [55.30, 60.00) s:
  mass x [0.861296, 0.981169, 1.11128, 1.19447]
  CoM XYZ mm [-4.05623, -2.17658, -2.20144]
  mu_s [0.785466, 0.794338, 0.598343, 0.547743]
  mu_d [0.52258, 0.526209, 0.598343, 0.547743]
  Kp [steer,rear] [9.01218, 0]; Kd [0.195789, 0]
  spawn roll -0.0596743 rad; rate -0.183025 rad/s
  gyro bias XYZ rad/s [0.000484342, -0.000219137, -0.000125366]
  gravity bias YZ [-0.00415205, 0.00169506]
Six significant digits shown; full precision: sampled_dr_values.json.
```

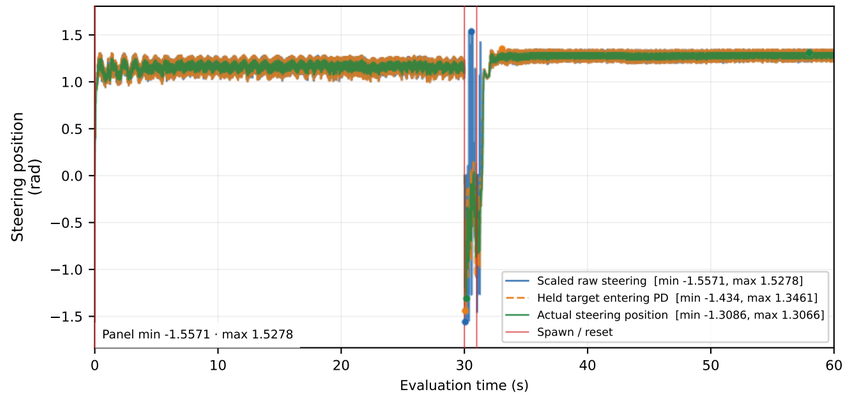
```
SAMPLED DR: env 22; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.46) s:
  mass x [1.1774, 1.00934, 0.826284, 0.927424]
  CoM XYZ mm [3.58417, -4.15441, 4.85365]
  mu_s [0.704749, 0.631146, 0.736077, 0.538434]
  mu_d [0.541926, 0.631146, 0.699455, 0.538434]
  Kp [steer,rear] [9.06763, 0]; Kd [0.202224, 0]
  spawn roll 0.0374538 rad; rate 0.0326712 rad/s
  gyro bias XYZ rad/s [-0.000140744, -0.000192308, 0.000153964]
  gravity bias YZ [-0.000118989, 0.00169111]
E2 [1.46, 31.46) s:
  mass x [0.836772, 1.01964, 1.09406, 1.12996]
  CoM XYZ mm [-2.6505, -3.9608, -3.50786]
  mu_s [0.727426, 0.757579, 0.733515, 0.568073]
  mu_d [0.606358, 0.70921, 0.558392, 0.568073]
  Kp [steer,rear] [9.24874, 0]; Kd [0.218677, 0]
  spawn roll -0.0268408 rad; rate 0.194205 rad/s
  gyro bias XYZ rad/s [-6.24926e-05, 7.9838e-05, -0.000160236]
  gravity bias YZ [-0.00274159, -0.000387806]
E3 [31.46, 32.34) s:
  mass x [0.890461, 1.13429, 0.828668, 1.07681]
  CoM XYZ mm [4.4716, -2.65451, -3.88171]
  mu_s [0.700521, 0.69316, 0.749678, 0.723429]
  mu_d [0.641329, 0.69316, 0.741429, 0.723429]
  Kp [steer,rear] [10.2565, 0]; Kd [0.217698, 0]
  spawn roll -0.0679824 rad; rate -0.0581301 rad/s
  gyro bias XYZ rad/s [1.34444e-05, 1.75308e-06, -1.70683e-05]
  gravity bias YZ [-0.000319099, -0.000625235]
E4 [32.34, 32.72) s:
  mass x [0.875501, 1.04263, 0.881064, 1.04699]
  CoM XYZ mm [-1.33345, -4.39802, 3.62697]
  mu_s [0.633024, 0.538434, 0.757579, 0.704749]
  mu_d [0.568172, 0.538434, 0.70921, 0.541926]
  Kp [steer,rear] [9.25831, 0]; Kd [0.198036, 0]
  spawn roll 0.0501151 rad; rate 0.207299 rad/s
  gyro bias XYZ rad/s [-0.000203632, 0.000105006, 0.000199391]
  gravity bias YZ [0.00161353, 0.00366085]
E5 [32.72, 33.02) s:
  mass x [1.14551, 1.11546, 0.866295, 0.830485]
  CoM XYZ mm [0.0241074, 3.01149, 0.832467]
  mu_s [0.733515, 0.559972, 0.633024, 0.736077]
  mu_d [0.558392, 0.502209, 0.568172, 0.699455]
  Kp [steer,rear] [9.22348, 0]; Kd [0.193362, 0]
  spawn roll -0.0414557 rad; rate -0.127543 rad/s
  gyro bias XYZ rad/s [-0.000168263, 0.000126677, -0.000223143]
  gravity bias YZ [-0.00219662, 0.000639729]
E6 [33.02, 60.00) s:
  mass x [0.873044, 0.892759, 1.0757, 0.971105]
  CoM XYZ mm [1.30932, -4.6614, -3.35964]
  mu_s [0.69316, 0.700521, 0.723429, 0.736077]
  mu_d [0.69316, 0.641329, 0.723429, 0.699455]
  Kp [steer,rear] [10.374, 0]; Kd [0.18768, 0]
  spawn roll -0.0733841 rad; rate -0.114182 rad/s
  gyro bias XYZ rad/s [0.000219676, 0.000425842, 0.000271977]
  gravity bias YZ [-0.00149297, -0.00327126]
Six significant digits shown; full precision: sampled_dr_values.json.
```

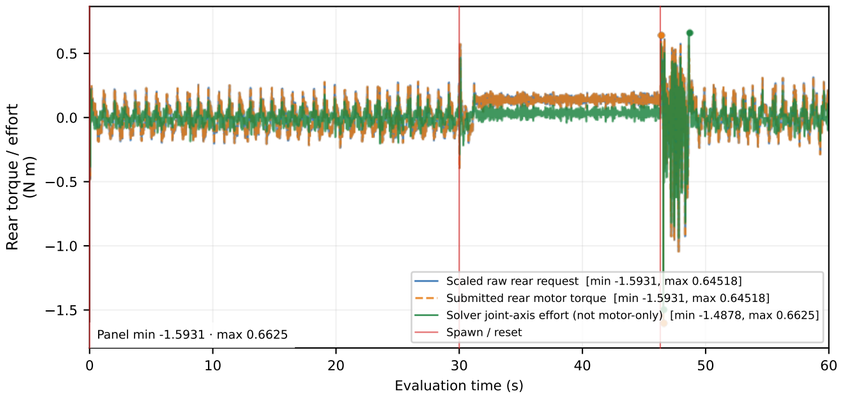
```
SAMPLED DR: env 22; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.1774, 1.00934, 0.826284, 0.927424]
  CoM XYZ mm [3.58417, -4.15441, 4.85365]
  mu_s [0.704749, 0.631146, 0.736077, 0.538434]
  mu_d [0.541926, 0.631146, 0.699455, 0.538434]
  Kp [steer,rear] [9.06763, 0]; Kd [0.202224, 0]
  spawn roll 0.0374538 rad; rate 0.0326712 rad/s
  gyro bias XYZ rad/s [-0.000140744, -0.000192308, 0.000153964]
  gravity bias YZ [-0.000118989, 0.00169111]
E2 [30.00, 60.00) s:
  mass x [1.10383, 1.04039, 1.03433, 1.08747]
  CoM XYZ mm [0.44504, -1.15206, 3.22108]
  mu_s [0.749678, 0.748289, 0.748289, 0.749678]
  mu_d [0.741429, 0.689499, 0.689499, 0.741429]
  Kp [steer,rear] [10.2047, 0]; Kd [0.202195, 0]
  spawn roll -0.069652 rad; rate -0.0482418 rad/s
  gyro bias XYZ rad/s [0.000186282, 2.88473e-05, -0.000391111]
  gravity bias YZ [0.0013609, -0.00096558]
Six significant digits shown; full precision: sampled_dr_values.json.
```

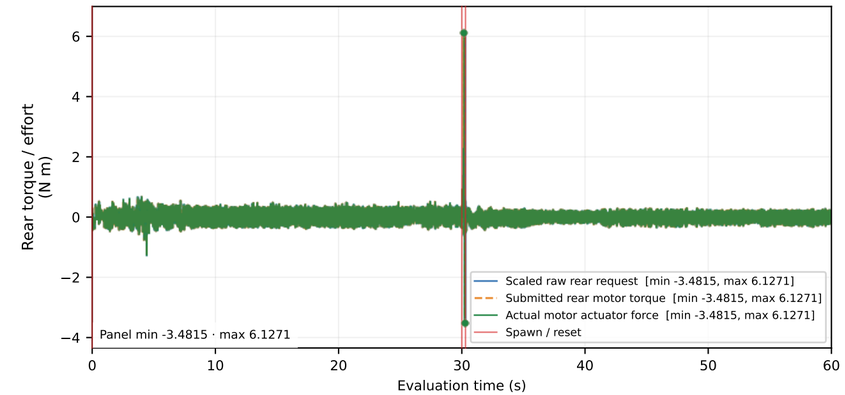
model_2500 · Demand versus actuation · env 23 · full 60 s

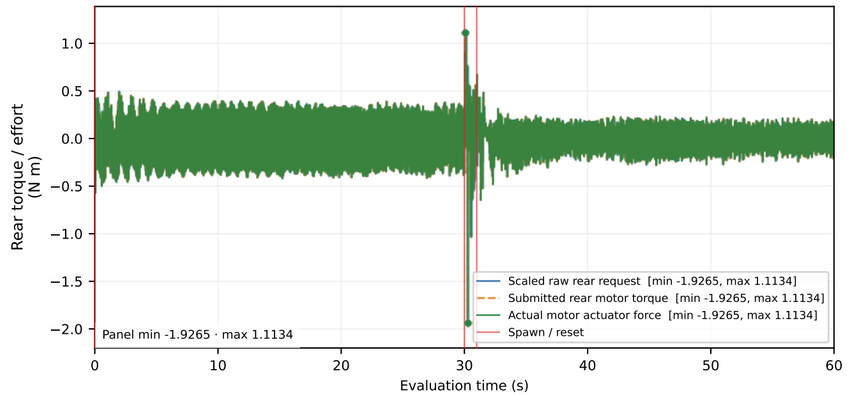












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 23; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.902276, 1.07803, 1.00334, 0.809615]
  CoM XYZ mm [3.325, 2.94543, -3.45263]
  mu_s [0.703711, 0.531594, 0.713129, 0.713129]
  mu_d [0.703711, 0.531594, 0.713129, 0.713129]
  Kp [steer,rear] [9.33442, 0]; Kd [0.190002, 0]
  spawn roll 0.0350462 rad; rate -0.216377 rad/s
  gyro bias XYZ rad/s [0.00011978, -9.29031e-05, 0.000219528]
  gravity bias YZ [0.00326947, 0.00403598]
E2 [30.00, 46.32) s:
  mass x [0.960775, 1.19447, 1.09524, 0.929989]
  CoM XYZ mm [1.72573, -4.97271, -3.64296]
  mu_s [0.539956, 0.787792, 0.647391, 0.584424]
  mu_d [0.539956, 0.617134, 0.647391, 0.584424]
  Kp [steer,rear] [10.5864, 0]; Kd [0.219723, 0]
  spawn roll 0.00635613 rad; rate 0.130233 rad/s
  gyro bias XYZ rad/s [0.000137682, 0.000124554, 0.000185817]
  gravity bias YZ [-9.12121e-06, -0.0011008]
E3 [46.32, 60.00) s:
  mass x [1.07932, 0.846165, 1.19697, 1.03142]
  CoM XYZ mm [-0.91359, 2.53814, -1.48316]
  mu_s [0.602694, 0.726175, 0.560535, 0.659445]
  mu_d [0.602694, 0.558574, 0.560535, 0.547619]
  Kp [steer,rear] [9.03699, 0]; Kd [0.194668, 0]
  spawn roll -0.0463249 rad; rate -0.0693574 rad/s
  gyro bias XYZ rad/s [-0.000236129, 4.4613e-05, -6.95118e-05]
  gravity bias YZ [-0.00167178, -0.000764033]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 23; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.818957, 1.00729, 0.949943, 0.891471]
  CoM XYZ mm [4.41723, 0.945015, -2.21149]
  mu_s [0.568073, 0.757579, 0.559972, 0.538434]
  mu_d [0.568073, 0.70921, 0.502209, 0.538434]
  Kp [steer,rear] [9.0213, 0]; Kd [0.181029, 0]
  spawn roll 0.0108942 rad; rate 0.249432 rad/s
  gyro bias XYZ rad/s [0.000368914, -1.73389e-05, 0.000229485]
  gravity bias YZ [-0.000507162, -0.00263101]
E2 [30.00, 30.30) s:
  mass x [1.09747, 1.09297, 0.987042, 0.984079]
  CoM XYZ mm [4.1462, 4.50607, -3.18988]
  mu_s [0.538976, 0.640867, 0.791209, 0.757579]
  mu_d [0.538976, 0.556841, 0.767936, 0.70921]
  Kp [steer,rear] [9.91003, 0]; Kd [0.187922, 0]
  spawn roll -0.0730179 rad; rate -0.233558 rad/s
  gyro bias XYZ rad/s [9.01731e-06, 0.0001623, 6.279e-05]
  gravity bias YZ [0.00214148, -0.00207834]
E3 [30.30, 60.00) s:
  mass x [0.939329, 1.19123, 0.965854, 1.16404]
  CoM XYZ mm [3.06904, -0.920233, 2.4337]
  mu_s [0.723429, 0.568073, 0.71155, 0.748289]
  mu_d [0.723429, 0.568073, 0.71155, 0.689499]
  Kp [steer,rear] [9.03876, 0]; Kd [0.218117, 0]
  spawn roll -0.00209098 rad; rate 0.0701048 rad/s
  gyro bias XYZ rad/s [-0.000131661, 0.000121779, -0.000318406]
  gravity bias YZ [-0.00128338, 0.000232989]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 23; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.818957, 1.00729, 0.949943, 0.891471]
  CoM XYZ mm [4.41723, 0.945015, -2.21149]
  mu_s [0.568073, 0.757579, 0.559972, 0.538434]
  mu_d [0.568073, 0.70921, 0.502209, 0.538434]
  Kp [steer,rear] [9.0213, 0]; Kd [0.181029, 0]
  spawn roll 0.0108942 rad; rate 0.249432 rad/s
  gyro bias XYZ rad/s [0.000368914, -1.73389e-05, 0.000229485]
  gravity bias YZ [-0.000507162, -0.00263101]
E2 [30.00, 31.00) s:
  mass x [1.09747, 1.09297, 0.987042, 0.984079]
  CoM XYZ mm [4.1462, 4.50607, -3.18988]
  mu_s [0.538976, 0.640867, 0.791209, 0.757579]
  mu_d [0.538976, 0.556841, 0.767936, 0.70921]
  Kp [steer,rear] [9.91003, 0]; Kd [0.187922, 0]
  spawn roll -0.0730179 rad; rate -0.233558 rad/s
  gyro bias XYZ rad/s [9.01731e-06, 0.0001623, 6.279e-05]
  gravity bias YZ [0.00214148, -0.00207834]
E3 [31.00, 60.00) s:
  mass x [0.918573, 0.834252, 0.82717, 1.00434]
  CoM XYZ mm [1.36049, -2.84484, 2.19587]
  mu_s [0.611239, 0.640867, 0.69316, 0.559972]
  mu_d [0.611239, 0.556841, 0.69316, 0.502209]
  Kp [steer,rear] [10.1131, 0]; Kd [0.189644, 0]
  spawn roll -0.0488518 rad; rate -0.186283 rad/s
  gyro bias XYZ rad/s [-0.000249822, 0.000127333, 2.32506e-06]
  gravity bias YZ [0.00168369, -0.00131681]
Six significant digits shown; full precision: sampled_dr_values.json.
```